

PolyFlop8

ISTQB Test Requirements Specification

Block Level & ISA Execution Verification

Revision: 1.1
Date: January 6, 2026

Based on PolyFlop8 Technical Reference Manual

1 REVISION HISTORY

Version	Description	Date
1.1	Updated TR-REG-01 to include 8-bit bus and bit slicing [7:5]. Added TR-RAM-03 for Computed Address Mode verification.	06.01.2026
1.0	Initial document creation	05.01.2026

2 HARDWARE BLOCK TEST REQUIREMENTS

This section defines the functional requirements for individual hardware components (Unit Testing).

2.1 ALU (Arithmetic Logic Unit)

Req ID	Requirement Description	Traceability
TR-ALU-01	Verify all arithmetic/logical operations (ADD, SUB, AND, XOR, etc.) defined in the Operations Table produce correct 8-bit results.	Table 2.3
TR-ALU-02	Verify flags (H, V, N, C, Z) are correctly asserted for each operation type according to the "Flags Affected" column.	Sec 2.2
TR-ALU-03	Verify <code>alu_mux_sel_a</code> correctly selects between <code>regfile_data_a ('0')</code> and <code>pc_data ('1')</code> .	Sec 2.2
TR-ALU-04	Verify <code>alu_mux_sel_b</code> correctly selects between <code>regfile_data_b ('0')</code> and <code>rs_imm_in ('1')</code> .	Sec 2.2
TR-ALU-05	Verify 2's complement handling in NEG and SUB operations, specifically checking Overflow (V) flag logic.	Table 2.3

2.2 Instruction Decoder

Req ID	Requirement Description	Traceability
TR-DEC-01	Verify <code>opcode</code> output correctly extracts bits [15:11] from the 16-bit <code>decoder_data_in</code> .	Table 2, Slice 1
TR-DEC-02	Verify <code>RD</code> output correctly extracts bits [10:8] representing the destination register.	Table 2, Slice 2
TR-DEC-03	Verify <code>RS_IMM</code> output correctly extracts bits [7:0] for use as either Source Register or Immediate value.	Table 2, Slice 3
TR-DEC-04	Verify <code>Jump11bit</code> output correctly extracts bits [10:0] for absolute jump addresses.	Table 2, Slice 4

2.3 Program Counter (PC)

Req ID	Requirement Description	Traceability
TR-PC-01	Verify Fetch Mode (<code>pc_src="00"</code>): PC increments by 1 on each clock cycle.	Table 4.3
TR-PC-02	Verify Branch Mode (<code>pc_src="01"</code>): PC updates to $(PC + 1) + \text{sign_extend}(\text{alu_out})$.	Table 4.3
TR-PC-03	Verify Indirect/Ret Mode (<code>pc_src="10"</code>): PC loads address from <code>ram_data</code> .	Table 4.3
TR-PC-04	Verify Jump Mode (<code>pc_src="11"</code>): PC loads absolute address from <code>jump_abs_11bit</code> .	Table 4.3
TR-PC-05	Verify <code>pc_en</code> signal effectively stalls the PC (no change) when low.	Sec 4.2

2.4 Register File

Req ID	Requirement Description	Traceability
TR-REG-01	Verify asynchronous read on <code>data_a</code> and <code>data_b</code> based on addresses <code>addr_a</code> and <code>addr_b</code> , and ensure the Register File correctly accepts an 8-bit bus and slices bits [7:5] for register selection.	Sec 5
TR-REG-02	Verify synchronous write to register at <code>addr_w</code> only when <code>we</code> is high.	Sec 5.3
TR-REG-03	Verify <code>data_r7x</code> acts as a dedicated hardwired output for Register 7.	Sec 5.2
TR-REG-04	Verify Write Mux (<code>rf_src_sel</code>) correctly selects sources: ALU, RAM, Immediate, or I/O.	Sec 5.2

2.5 Data RAM

Req ID	Requirement Description	Traceability
TR-RAM-01	Verify asynchronous read returns the value stored at the selected address without requiring a clock edge.	DataRAM, Read Logic
TR-RAM-02	Verify synchronous write updates memory only on rising clock edge when <code>mem_we = '1'</code> .	DataRAM, Write Process
TR-RAM-03	Verify the Computed Address Mode , where the RAM address is driven by the ALU output, and ensure correct data is read/written at the computed address.	DataRAM Addr MUX
TR-RAM-04	Verify Data MUX selects correct write data source: Reg B, PC, or ALU output.	DataRAM Data MUX
TR-RAM-05	Verify memory boundary behavior for lowest (0x00) and highest (0xFF) addresses.	DataRAM Address Range

2.6 Status Register (SREG)

Req ID	Requirement Description	Traceability
TR-SREG-01	Verify flags H, V, N, Z, C are correctly latched from ALU outputs when <code>sreg_we = '1'</code> and <code>sreg_src = '0'</code> .	StatusReg ALU Mode
TR-SREG-02	Verify full 8-bit SREG is loaded from RegFile when <code>sreg_src = '1'</code> (POP/OUT behavior).	StatusReg RegFile Mode
TR-SREG-03	Verify Sign flag (S) is computed as <code>N xor V</code> .	StatusReg Bit Mapping
TR-SREG-04	Verify bits I and T retain previous values during ALU flag updates.	StatusReg Latch Logic
TR-SREG-05	Verify asynchronous reset clears entire Status Register.	StatusReg Reset

2.7 Program ROM (Instruction Memory)

Req ID	Requirement Description	Traceability
TR-ROM-01	Verify instruction fetch returns correct 16-bit word for given 11-bit PC address.	ProgPROM Read Logic
TR-ROM-02	Verify ROM read is asynchronous and does not depend on clock edge.	ProgPROM Architecture
TR-ROM-03	Verify correct instruction encoding for LDI, ADD, SUB, and JMP instructions.	ROM Program Image
TR-ROM-04	Verify JMP instruction correctly maps 11-bit address field into instruction word.	ProgPROM Opcode Format
TR-ROM-05	Verify undefined addresses return NOP-equivalent instruction.	ROM Default Case

2.8 I/O Unit

Req ID	Requirement Description	Traceability
TR-IO-01	Verify OUT instruction writes register value to output port when <code>io_we = '1'</code> .	IO_Unit Write Logic
TR-IO-02	Verify correct output port selection based on <code>io_addr_in</code> .	IO Address Decode
TR-IO-03	Verify IN instruction reads external pin values into register via <code>io_data_in</code> .	IO Read Logic
TR-IO-04	Verify output port retains last written value until updated or reset.	IO Output Register
TR-IO-05	Verify reset clears all output port registers to zero.	IO Reset Logic

2.9 Control Unit

Req ID	Requirement Description	Traceability
TR-CU-01	Verify FSM transitions correctly through FETCH → DE-CODE → EXECUTE states.	ControlUnit FSM
TR-CU-02	Verify instruction fetch enables IR write only during FETCH state.	IR Enable Logic
TR-CU-03	Verify correct ALU control signals are generated for arithmetic instructions.	ALU Control Mapping
TR-CU-04	Verify memory write enable is asserted only for STORE instructions.	DRAM Control
TR-CU-05	Verify register file write-back occurs only for instructions producing results.	RegFile WE Logic
TR-CU-06	Verify PC update source is correctly selected for sequential, branch, and jump instructions.	PC Control Logic
TR-CU-07	Verify default control outputs prevent unintended writes (NOP safety).	Default Signal Values

3 ISA EXECUTION TEST REQUIREMENTS

This section defines requirements for System Integration Testing, verifying the CPU executes code correctly (Equivalence Partitioning & Boundary Value Analysis).

3.1 Arithmetic & Logic Instructions

Req ID	Test Requirement Description	Instruction
TR-ISA-01	Verify ADD and ADC correctly sum two 8-bit registers (Equivalence Partitioning).	ADD, ADC
TR-ISA-02	Verify ADD sets the Carry Flag (C) when result exceeds 255 (Boundary Value).	ADD
TR-ISA-03	Verify ADD sets the Overflow Flag (V) when signed 2's complement addition wraps incorrectly.	ADD
TR-ISA-04	Verify SUB and SBB correctly subtract two 8-bit registers (checking Reg-Reg and Reg-Imm).	SUB, SBB
TR-ISA-05	Verify SUB sets the Zero Flag (Z) only when the result is exactly 0.	SUB, CMP
TR-ISA-06	Verify Logical Operations (AND, OR, XOR) correctly manipulate bits and clear/set flags.	AND, OR
TR-ISA-07	Verify INC and DEC increment/decrement target without affecting Carry flag (if specified).	INC, DEC

3.2 Data Transfer Instructions

Req ID	Test Requirement Description	Instruction
TR-ISA-08	Verify Immediate Load : Loading a constant (e.g., LDI R1, 0xFF) results in exact value in destination.	LDI
TR-ISA-09	Verify Register Move : Copying data from Rs to Rd preserves value; Source unchanged.	MOV
TR-ISA-10	Verify Direct Store : Storing register to memory address (ST [Addr], R1) writes correct data.	STS
TR-ISA-11	Verify Direct Load : Loading from a valid memory address retrieves previously written data.	LDS
TR-ISA-12	Verify Indirect Addressing : Accessing memory via pointer (LD R1, [R7]) accesses address in R7.	LD/ST

3.3 Program Flow Control

Req ID	Test Requirement Description	Instruction
TR-ISA-13	Verify Unconditional Jump : JMP updates PC to target immediately, regardless of flags.	RJMP
TR-ISA-14	Verify Conditional Branch (Taken) : BR updates PC only if condition (e.g., Z=1) is met.	BRBS/BRBC
TR-ISA-15	Verify Conditional Branch (Not Taken) : BR proceeds to PC + 1 if condition is NOT met.	BRBS/BRBC
TR-ISA-16	Verify Call : RCALL pushes Return Address (PC + 1) onto Stack before jumping.	RCALL
TR-ISA-17	Verify Return : RET pops top value from Stack into PC, resuming execution.	RET

3.4 Stack & I/O Instructions

Req ID	Test Requirement Description	Instruction
TR-ISA-18	Verify PUSH/POP : Pushing/Popping correctly updates Stack Pointer and memory.	PUSH/POP
TR-ISA-19	Verify I/O OUT : OUT instruction writes register value to specified Output Port latch.	OUT
TR-ISA-20	Verify I/O IN : IN instruction reads current logic level of Input Port pins into register.	IN